

EX NO:2	Chebyshev IIR Filter
DATE:	2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

Code:

```
clc; clear;
```

```
% Specifications
```

```
Wp = 0.4; % Passband edge (normalized, 0 to 1  $\rightarrow \pi$  rad/sample)
```

```
Ws = 0.55; % Stopband edge (normalized)
```

```
Rp = 1; % Passband ripple (dB)
```

```
As = 40; % Stopband attenuation (dB)
```

```
% Filter design
```

```
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);
```

```
...% Group Delay
```

```
figure;
```

```
grpdelay(b, a);
```

```
title('Group Delay');
```

```
% Pole-Zero Plot
```

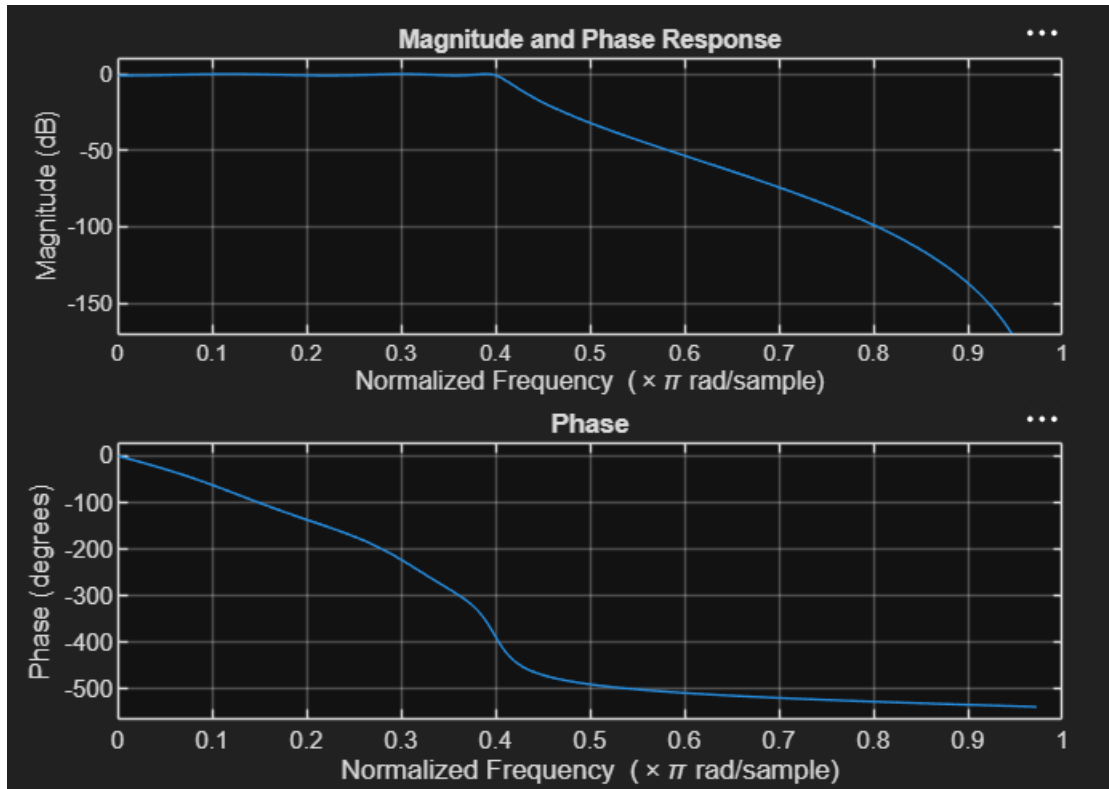
```
figure;
```

```
zplane(b, a);
```

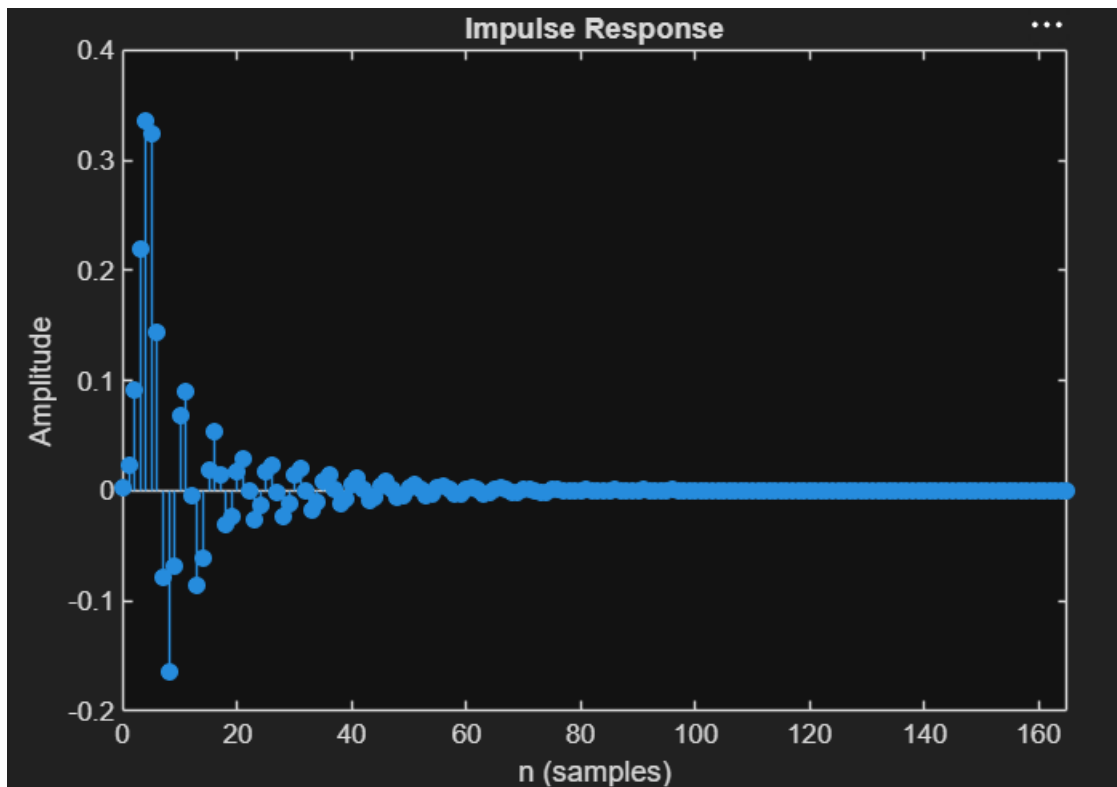
```
title('Pole-Zero Plot');
```

Output:

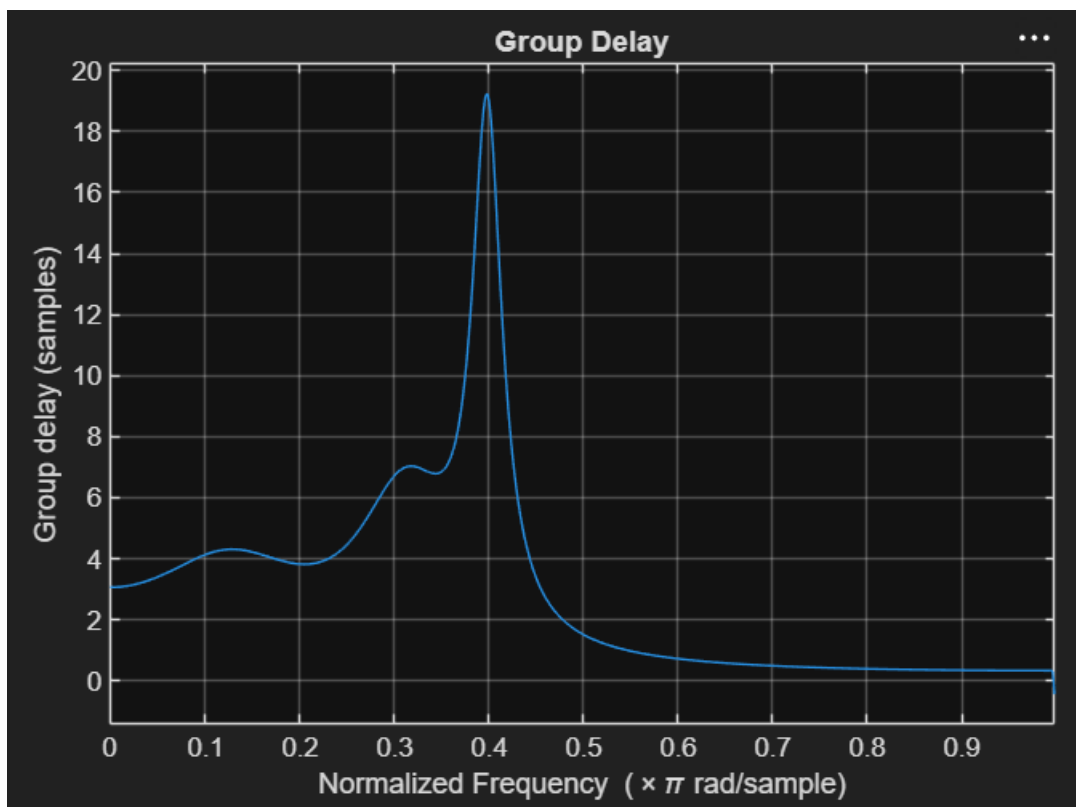
Observation 1:



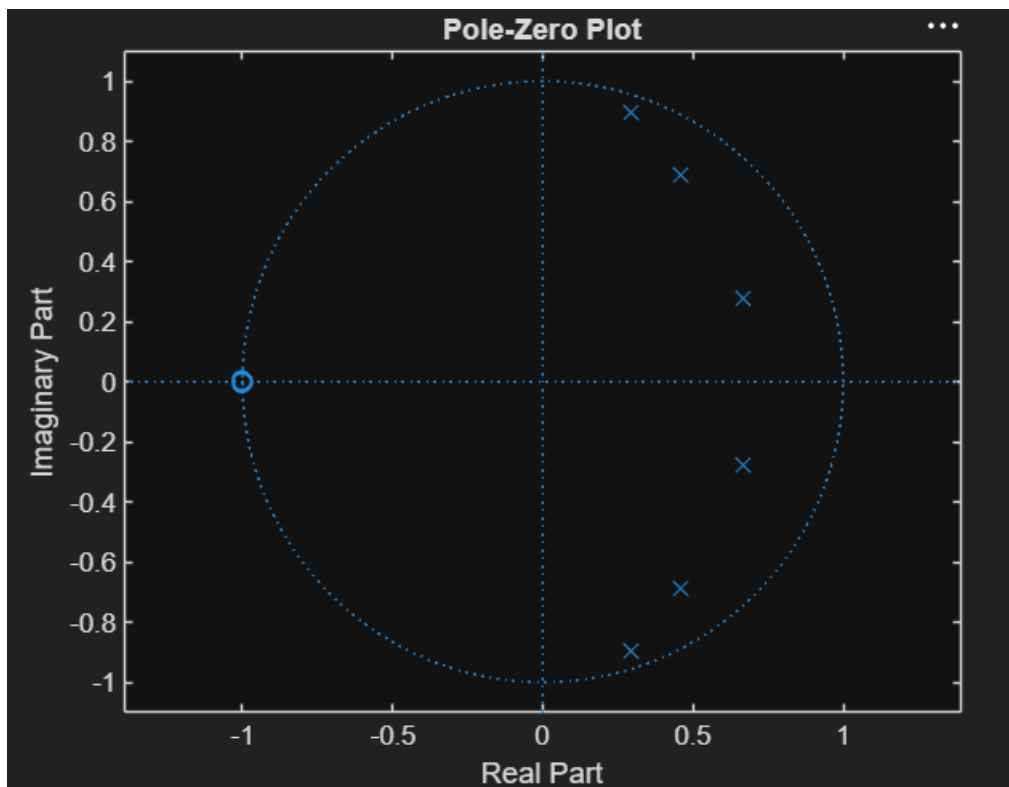
Observation 2:



Observation 3:



Observation 4:



EX NO:2	Chebyshev IIR Filter
DATE:	2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

Code:

```

clc;

clear;

close all;


% Specifications

Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge (normalized, 0–1 →  $\pi$  rad/sample)
Ws = 0.35; % Stopband Edge (normalized)


...

% Pole-Zero Plot

figure;

zplane(b, a);

title('Pole-Zero Plot');


% Display results

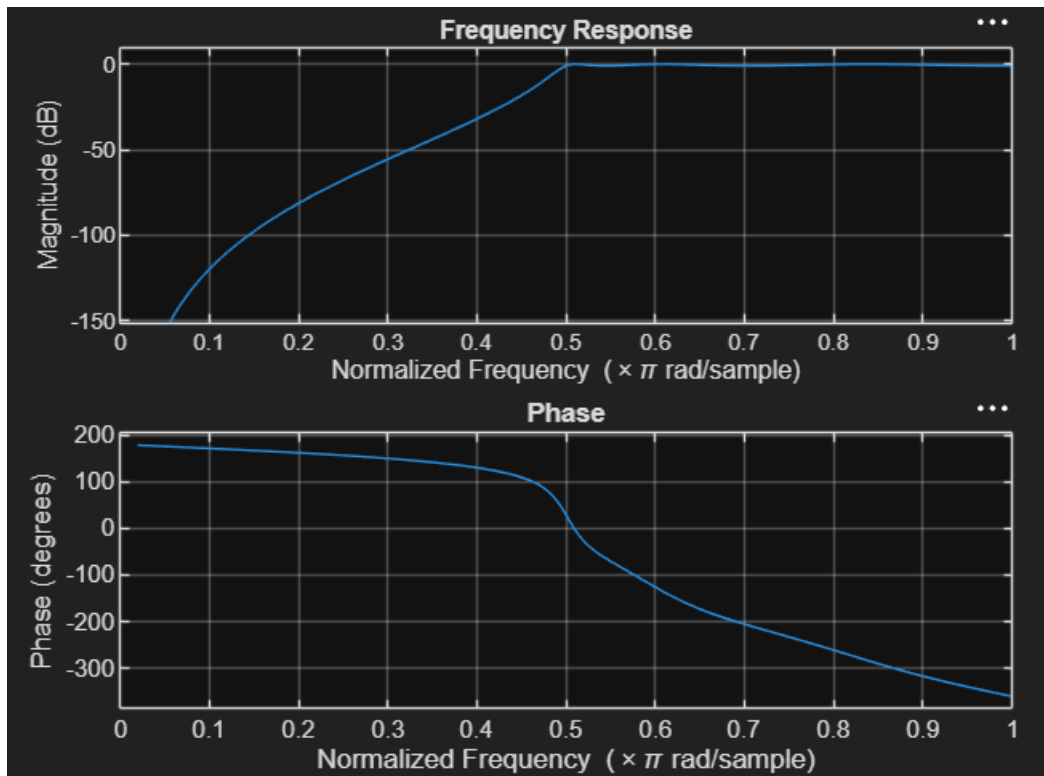
fprintf('Filter Order = %d\n', n);

fprintf('Cutoff Frequency = %.4f (normalized)\n', Wn);

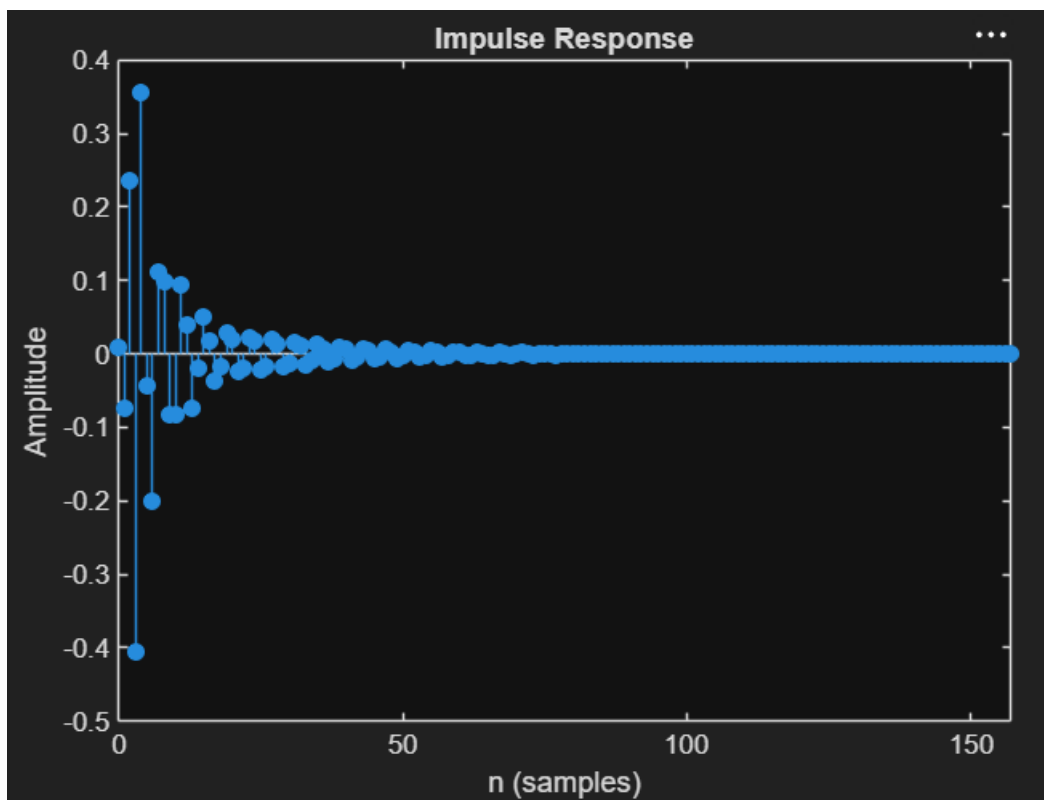
```

Output:

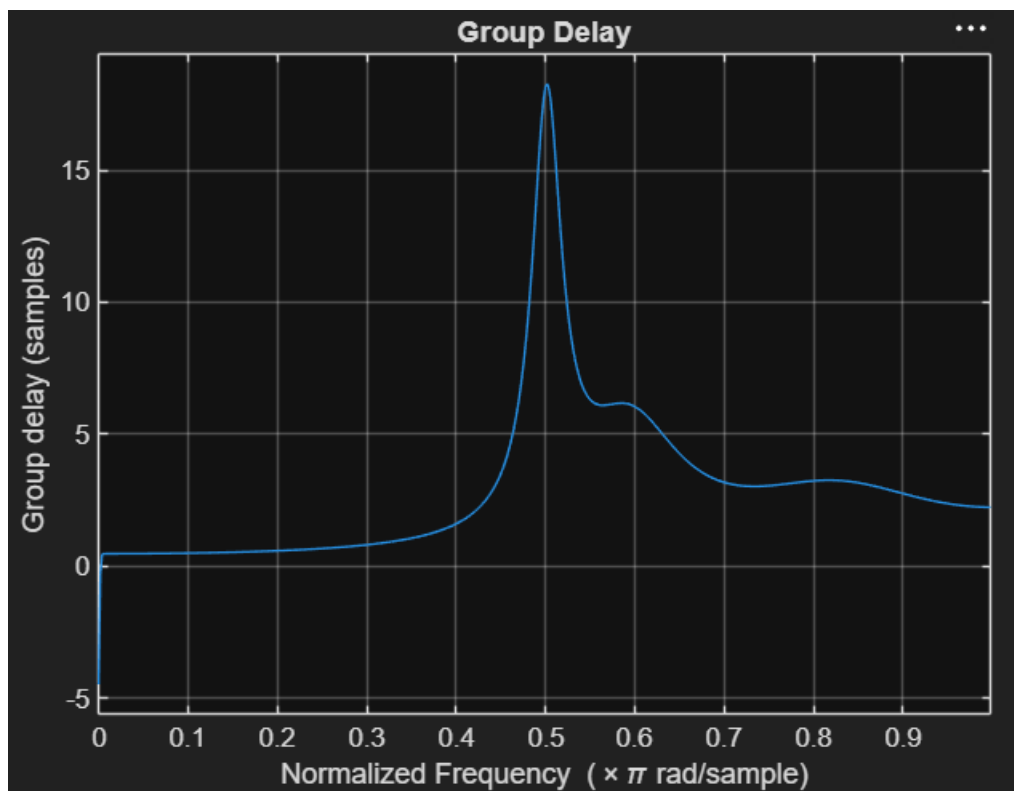
Observation 1:



Observation 2:



Observation 3:



Observation 4:

